

SidewalkPilot Steering Model Report

Combined current-label offline evaluation

Generated 2026-05-10 23:32:13. All available SidewalkPilot steering checkpoints are evaluated on the same active correction set: 2,224 labeled images from steering_corrections.json. Active checkpoint naming is SidewalkPilot-vX.Y.pth for final checkpoints and SidewalkPilot-vX.Yb.pth for best checkpoints.

Best offline checkpoint on this label set: 2.4b (SidewalkPilot-v2.4b.pth) with MAE 3.265, median AE 2.389, score 98.186%. Second: 2.4 (SidewalkPilot-v2.4.pth) with MAE 3.284.

Series 1 uses raw BGR and output scale 86. Series 2 uses output scale 85; v2.0/v2.0b use legacy HSV/CLAHE preprocessing, while v2.1 and newer use raw BGR. These are offline same-dataset decision comparisons; field reliability still has to be proven on the car.

Active Correction Sources

Source	Count	Purpose
D0328	315	First dataset relabel, March 28
D0329	413	First dataset relabel, March 29
D0425	65	Street test
D0426	53	Curves and shadows
D0427	72	Curved curb
D0429	53	Driveway and shadow
D0502_12	154	Shadow fix set
D0502_19	156	Hard turns, curb hugging, smoothness
D0503	159	Harsh sidewalk
D0506	24	8pm sidewalk failure set
D0510	760	v2.3 field-failure runs: turns, road-right driving, driveways

Combined Model Ranking

Rank	Model	Checkpoint filename	Prep	MAE	Med	Max	Signed	<=5	<=10	Score
1	2.4b	SidewalkPilot-v2.4b.pth	raw BGR	3.27	2.39	37.46	-0.04	1729/2224	2130/2224	98.186%
2	2.4	SidewalkPilot-v2.4.pth	raw BGR	3.28	2.46	35.27	-0.23	1724/2224	2130/2224	98.176%
3	2.3	SidewalkPilot-v2.3.pth	raw BGR	3.30	1.93	74.41	0.01	1768/2224	2102/2224	98.164%
4	2.3b	SidewalkPilot-v2.3b.pth	raw BGR	3.41	2.05	72.66	0.41	1743/2224	2107/2224	98.104%
5	2.2b	SidewalkPilot-v2.2b.pth	raw BGR	6.68	4.23	81.60	-0.46	1254/2224	1785/2224	96.289%
6	2.2	SidewalkPilot-v2.2.pth	raw BGR	6.68	4.26	82.03	-0.22	1238/2224	1802/2224	96.289%

Rank	Model	Checkpoint filename	Prep	MAE	Med	Max	Signed	<=5	<=10	Score
7	2.1b	SidewalkPilot-v2.1b.pth	raw BGR	10.75	5.79	155.58	-1.12	978/2224	1489/2224	94.029%
8	2.1	SidewalkPilot-v2.1.pth	raw BGR	10.80	5.83	156.37	-1.15	981/2224	1481/2224	93.998%
9	2.0	SidewalkPilot-v2.0.pth	HSV/CLAHE -> BGR	11.12	5.09	149.97	-2.63	1068/2224	1441/2224	93.820%
10	2.0b	SidewalkPilot-v2.0b.pth	HSV/CLAHE -> BGR	11.13	5.07	149.49	-2.62	1065/2224	1436/2224	93.817%
11	1.9b	SidewalkPilot-v1.9b.pth	raw BGR	12.94	6.60	164.82	-2.06	979/2224	1333/2224	92.812%
12	1.9	SidewalkPilot-v1.9.pth	raw BGR	12.98	6.63	164.79	-2.06	966/2224	1332/2224	92.787%
13	1.8b	SidewalkPilot-v1.8b.pth	raw BGR	15.41	9.36	170.47	-0.70	794/2224	1142/2224	91.440%
14	1.8	SidewalkPilot-v1.8.pth	raw BGR	15.47	9.46	170.49	-0.65	796/2224	1144/2224	91.405%
15	1.7b	SidewalkPilot-v1.7b.pth	raw BGR	15.54	9.54	161.47	-0.87	769/2224	1147/2224	91.366%
16	1.7	SidewalkPilot-v1.7.pth	raw BGR	15.82	9.67	160.93	-0.48	745/2224	1132/2224	91.212%
17	1.4	SidewalkPilot-v1.4.pth	raw BGR	16.81	10.19	160.98	-3.99	676/2224	1100/2224	90.661%
18	1.5	SidewalkPilot-v1.5.pth	raw BGR	16.82	10.59	159.65	-0.89	667/2224	1061/2224	90.654%
19	1.5b	SidewalkPilot-v1.5b.pth	raw BGR	16.82	10.59	159.65	-0.89	667/2224	1061/2224	90.654%
20	1.4b	SidewalkPilot-v1.4b.pth	raw BGR	17.04	10.07	160.94	-4.15	685/2224	1107/2224	90.532%
21	1.6b	SidewalkPilot-v1.6b.pth	raw BGR	17.41	11.04	167.80	-0.01	629/2224	1041/2224	90.326%
22	1.6	SidewalkPilot-v1.6.pth	raw BGR	17.53	11.22	168.37	0.21	630/2224	1029/2224	90.262%
23	1.3b	SidewalkPilot-v1.3b.pth	raw BGR	18.48	11.25	161.35	-3.65	580/2224	1024/2224	89.732%
24	1.3	SidewalkPilot-v1.3.pth	raw BGR	18.55	11.81	161.00	-3.43	556/2224	982/2224	89.696%
25	1.2b	SidewalkPilot-v1.2b.pth	raw BGR	18.63	11.85	161.00	-3.73	584/2224	1003/2224	89.648%
26	1.2	SidewalkPilot-v1.2.pth	raw BGR	18.64	11.78	161.00	-3.55	582/2224	997/2224	89.642%
27	1.1	SidewalkPilot-v1.1.pth	raw BGR	20.85	13.42	163.76	-5.16	510/2224	888/2224	88.415%
28	1.1b	SidewalkPilot-v1.1b.pth	raw BGR	20.85	13.42	163.76	-5.16	510/2224	888/2224	88.415%
29	1.0b	SidewalkPilot-v1.0b.pth	raw BGR	25.07	17.46	163.08	-7.09	351/2224	667/2224	86.071%
30	1.0	SidewalkPilot-v1.0.pth	raw BGR	25.55	18.45	161.51	-7.45	332/2224	647/2224	85.804%

Chronological Model Growth

Model	Checkpoint filename	Series	Prep	Scale	MAE	Median	Signed	<=2	<=5	<=20	Pred mean
1.0	SidewalkPilot-v1.0.pth	1	raw BGR	86	25.55	18.45	-7.45	130	332	1184	88.87
1.0b	SidewalkPilot-v1.0b.pth	1	raw BGR	86	25.07	17.46	-7.09	135	351	1228	89.24
1.1	SidewalkPilot-v1.1.pth	1	raw BGR	86	20.85	13.42	-5.16	242	510	1416	91.16
1.1b	SidewalkPilot-v1.1b.pth	1	raw BGR	86	20.85	13.42	-5.16	242	510	1416	91.16
1.2	SidewalkPilot-v1.2.pth	1	raw BGR	86	18.64	11.78	-3.55	267	582	1519	92.77
1.2b	SidewalkPilot-v1.2b.pth	1	raw BGR	86	18.63	11.85	-3.73	275	584	1547	92.59
1.3	SidewalkPilot-v1.3.pth	1	raw BGR	86	18.55	11.81	-3.43	261	556	1544	92.89
1.3b	SidewalkPilot-v1.3b.pth	1	raw BGR	86	18.48	11.25	-3.65	265	580	1553	92.67
1.4	SidewalkPilot-v1.4.pth	1	raw BGR	86	16.81	10.19	-3.99	320	676	1620	92.33
1.4b	SidewalkPilot-v1.4b.pth	1	raw BGR	86	17.04	10.07	-4.15	337	685	1598	92.17
1.5	SidewalkPilot-v1.5.pth	1	raw BGR	86	16.82	10.59	-0.89	375	667	1572	95.43
1.5b	SidewalkPilot-v1.5b.pth	1	raw BGR	86	16.82	10.59	-0.89	375	667	1572	95.43
1.6	SidewalkPilot-v1.6.pth	1	raw BGR	86	17.53	11.22	0.21	316	630	1556	96.53
1.6b	SidewalkPilot-v1.6b.pth	1	raw BGR	86	17.41	11.04	-0.01	318	629	1563	96.31
1.7	SidewalkPilot-v1.7.pth	1	raw BGR	86	15.82	9.67	-0.48	415	745	1616	95.84
1.7b	SidewalkPilot-v1.7b.pth	1	raw BGR	86	15.54	9.54	-0.87	410	769	1635	95.45
1.8	SidewalkPilot-v1.8.pth	1	raw BGR	86	15.47	9.46	-0.65	479	796	1645	95.67
1.8b	SidewalkPilot-v1.8b.pth	1	raw BGR	86	15.41	9.36	-0.70	492	794	1649	95.62
1.9	SidewalkPilot-v1.9.pth	1	raw BGR	86	12.98	6.63	-2.06	584	966	1728	94.26
1.9b	SidewalkPilot-v1.9b.pth	1	raw BGR	86	12.94	6.60	-2.06	594	979	1734	94.26
2.0	SidewalkPilot-v2.0.pth	2	HSV/CLAHE -> BGR	85	11.12	5.09	-2.63	609	1068	1838	93.70
2.0b	SidewalkPilot-v2.0b.pth	2	HSV/CLAHE -> BGR	85	11.13	5.07	-2.62	600	1065	1838	93.70
2.1	SidewalkPilot-v2.1.pth	2	raw BGR	85	10.80	5.83	-1.15	477	981	1867	95.18
2.1b	SidewalkPilot-v2.1b.pth	2	raw BGR	85	10.75	5.79	-1.12	478	978	1867	95.20
2.2	SidewalkPilot-v2.2.pth	2	raw BGR	85	6.68	4.26	-0.22	594	1238	2094	96.10
2.2b	SidewalkPilot-v2.2b.pth	2	raw BGR	85	6.68	4.23	-0.46	612	1254	2094	95.86
2.3	SidewalkPilot-v2.3.pth	2	raw BGR	85	3.30	1.93	0.01	1130	1768	2194	96.33
2.3b	SidewalkPilot-v2.3b.pth	2	raw BGR	85	3.41	2.05	0.41	1088	1743	2195	96.73

Model	Checkpoint filename	Series	Prep	Scale	MAE	Median	Signed	<=2	<=5	<=20	Pred mean
2.4	SidewalkPilot-v2.4.pth	2	raw BGR	85	3.28	2.46	-0.23	958	1724	2218	96.09
2.4b	SidewalkPilot-v2.4b.pth	2	raw BGR	85	3.27	2.39	-0.04	953	1729	2219	96.28

Field-Case / Subset MAE

Lower is better. D0510 combines the three v2.3 field-failure runs added after v2.3 training.

Model	D0328	D0329	D0425	D0426	D0427	D0429	D0502_12	D0502_19	D0503	D0506	D0510
2.4b	3.16	2.92	4.11	4.04	4.08	2.51	4.62	3.58	2.78	2.47	3.13
2.4	3.14	2.96	4.09	3.97	4.12	2.48	4.66	3.60	2.85	2.48	3.15
2.3	2.39	2.51	3.30	3.04	3.46	2.54	3.23	3.07	2.38	3.71	4.42
2.3b	2.45	2.57	3.41	3.15	3.42	2.52	3.46	3.23	2.50	4.00	4.55
2.2b	3.81	3.56	5.46	4.97	5.61	4.38	5.67	4.72	3.50	2.62	11.45
2.2	3.84	3.62	5.42	4.70	5.66	4.39	5.68	4.74	3.53	2.50	11.41
2.1b	19.65	19.96	3.65	3.19	3.81	3.33	4.10	3.64	2.42	3.64	9.13
2.1	19.70	20.13	3.69	3.26	3.89	3.32	4.14	3.62	2.47	3.60	9.16
2.0	17.97	20.51	2.71	2.03	2.57	1.69	2.35	2.26	1.66	36.00	10.80
2.0b	18.09	20.54	2.69	2.02	2.56	1.70	2.32	2.25	1.66	35.79	10.77
1.9b	17.10	21.79	1.66	1.49	1.73	1.43	1.36	1.78	28.24	35.89	10.74
1.9	17.27	21.90	1.66	1.52	1.74	1.41	1.32	1.79	28.23	35.45	10.76
1.8b	17.43	20.92	1.52	1.19	1.42	0.97	1.08	22.58	28.02	37.04	14.20
1.8	17.40	20.92	1.51	1.18	1.42	0.96	1.08	22.62	28.07	37.18	14.37
1.7b	16.52	20.73	1.39	1.03	1.22	1.01	17.43	22.68	27.59	30.87	12.06
1.7	16.55	21.03	1.44	1.12	1.23	0.93	17.51	22.84	27.33	30.78	12.70
1.4	16.34	20.23	0.88	0.56	31.68	25.83	18.73	24.90	26.43	34.05	11.00
1.5	16.69	20.32	1.24	0.88	0.99	25.74	18.59	21.31	26.42	26.32	14.71
1.5b	16.69	20.32	1.24	0.88	0.99	25.74	18.59	21.31	26.42	26.32	14.71
1.4b	16.94	20.62	1.00	0.72	31.91	26.65	19.21	24.99	26.13	34.16	11.06
1.6b	17.01	20.01	1.24	0.90	1.10	26.22	19.02	22.90	27.36	29.09	15.73
1.6	16.84	20.09	1.26	0.84	1.10	26.32	18.91	22.95	27.36	29.10	16.11
1.3b	16.63	20.54	1.02	36.75	35.63	24.12	19.66	23.53	34.63	32.20	11.25
1.3	16.46	20.12	0.87	37.56	34.06	25.26	19.51	23.50	34.04	32.99	11.90
1.2b	16.39	19.86	0.93	38.42	32.11	25.46	21.01	23.36	32.64	36.64	12.33
1.2	16.38	19.88	0.87	39.03	31.76	25.38	20.86	23.31	32.51	36.89	12.41
1.1	17.12	20.33	1.32	41.55	36.60	33.44	21.42	26.71	33.09	29.72	16.38

Model	D0328	D0329	D0425	D0426	D0427	D0429	D0502_12	D0502_19	D0503	D0506	D0510
1.1b	17.12	20.33	1.32	41.55	36.60	33.44	21.42	26.71	33.09	29.72	16.38
1.0b	17.36	20.30	49.19	45.39	54.06	37.39	29.16	30.30	40.52	40.61	18.15
1.0	18.07	21.08	49.03	46.31	53.69	37.44	29.57	30.14	41.72	40.77	18.52

Prediction Distribution

Model	Pred min	P05	P25	Median	P75	P95	Pred max	Pred mean	Target mean
2.4b	5.00	57.77	82.46	93.29	103.57	164.71	175.00	96.28	96.32
2.4	5.00	57.47	82.19	93.08	103.48	164.71	175.00	96.09	96.32
2.3	5.00	58.41	83.26	93.64	103.24	165.00	175.00	96.33	96.32
2.3b	5.00	58.65	83.54	93.98	104.04	164.77	175.00	96.73	96.32
2.2b	5.00	51.20	81.50	93.35	104.85	167.64	175.00	95.86	96.32
2.2	5.00	51.21	81.67	93.66	105.09	167.06	175.00	96.10	96.32
2.1b	5.00	47.19	79.94	93.50	105.85	161.30	175.00	95.20	96.32
2.1	5.00	46.92	79.86	93.55	105.96	161.85	175.00	95.18	96.32
2.0	5.00	41.37	77.37	92.45	106.30	160.31	175.00	93.70	96.32
2.0b	5.00	41.27	77.43	92.44	106.38	160.19	175.00	93.70	96.32
1.9b	4.00	46.67	78.77	92.67	107.24	157.43	176.00	94.26	96.32
1.9	4.00	46.17	78.76	92.52	107.33	157.90	176.00	94.26	96.32
1.8b	4.00	48.09	80.15	94.64	109.16	154.91	176.00	95.62	96.32
1.8	4.00	48.29	80.11	94.60	109.19	155.60	176.00	95.67	96.32
1.7b	4.00	49.28	81.18	94.08	108.58	153.13	176.00	95.45	96.32
1.7	4.00	49.40	81.31	94.43	109.13	153.53	176.00	95.84	96.32
1.4	4.00	48.35	80.64	92.54	102.58	136.88	176.00	92.33	96.32
1.5	4.00	47.33	80.31	93.99	108.58	153.09	176.00	95.43	96.32
1.5b	4.00	47.33	80.31	93.99	108.58	153.09	176.00	95.43	96.32
1.4b	4.00	48.11	80.19	92.30	101.92	137.30	176.00	92.17	96.32
1.6b	4.00	48.24	79.75	94.62	109.73	154.52	176.00	96.31	96.32
1.6	4.00	48.57	80.01	94.65	109.88	154.93	176.00	96.53	96.32
1.3b	4.00	47.97	79.34	92.96	106.08	134.30	176.00	92.67	96.32
1.3	4.00	47.72	79.07	93.19	107.25	133.68	176.00	92.89	96.32
1.2b	4.00	47.74	78.51	93.35	106.81	137.57	176.00	92.59	96.32
1.2	4.00	47.80	78.33	93.45	107.11	136.18	176.00	92.77	96.32
1.1	4.00	44.01	75.58	90.81	106.23	141.57	176.00	91.16	96.32
1.1b	4.00	44.01	75.58	90.81	106.23	141.57	176.00	91.16	96.32

Model	Pred min	P05	P25	Median	P75	P95	Pred max	Pred mean	Target mean
1.0b	9.01	43.91	72.06	88.64	104.90	136.78	173.93	89.24	96.32
1.0	8.46	42.83	71.12	88.88	105.19	136.92	174.15	88.87	96.32

Prediction Buckets vs Ground

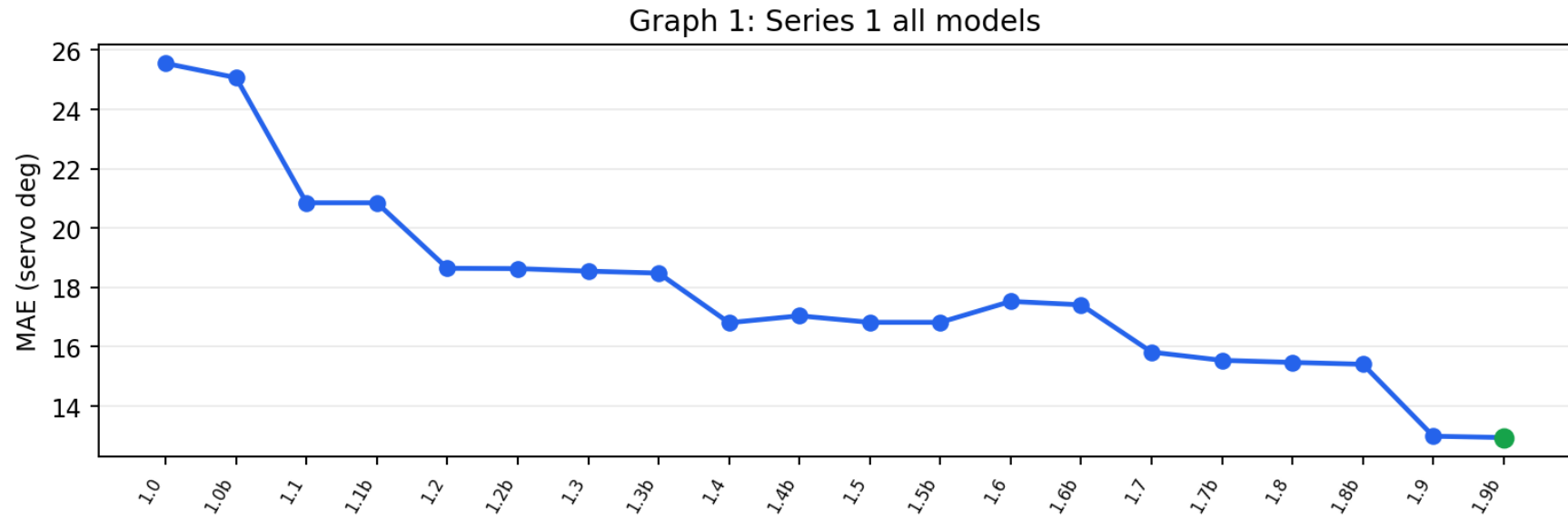
Bucket columns show prediction counts; exact/off-by-one compare prediction bucket against ground bucket.

Model	HL 0-45	L 45-75	SL 75-85	ST 85-95	SR 95-105	R 105-135	HR 135-180	Exact	Off by <=1
2.4b	76	200	390	543	507	296	212	71.2%	98.5%
2.4	76	209	389	552	498	292	208	70.8%	98.4%
2.3	73	181	380	578	515	292	205	74.3%	98.2%
2.3b	74	173	371	560	530	308	208	75.8%	98.2%
2.2b	94	252	371	492	466	330	219	58.5%	92.0%
2.2	94	244	366	488	473	342	217	58.9%	92.3%
2.1b	96	327	309	449	458	361	224	50.2%	84.9%
2.1	97	331	302	457	455	358	224	50.6%	84.8%
2.0	123	375	310	430	395	374	217	49.3%	84.8%
2.0b	123	373	310	433	393	373	219	49.6%	84.7%
1.9b	107	352	320	453	387	400	205	44.0%	81.9%
1.9	106	354	322	451	388	399	204	43.4%	81.8%
1.8b	99	332	278	426	404	481	204	40.0%	78.6%
1.8	99	332	276	424	404	479	210	40.1%	78.6%
1.7b	92	313	280	470	418	456	195	39.3%	79.6%
1.7	93	313	271	463	403	477	204	39.1%	78.9%
1.4	86	307	327	545	482	357	120	34.4%	75.4%
1.5	99	315	304	446	393	465	202	35.1%	76.9%
1.5b	99	315	304	446	393	465	202	35.1%	76.9%
1.4b	94	300	313	589	470	333	125	34.5%	75.0%
1.6b	92	331	290	420	413	441	237	34.2%	76.9%
1.6	91	339	283	419	405	445	242	33.5%	76.5%
1.3b	85	342	321	457	420	491	108	31.9%	74.0%

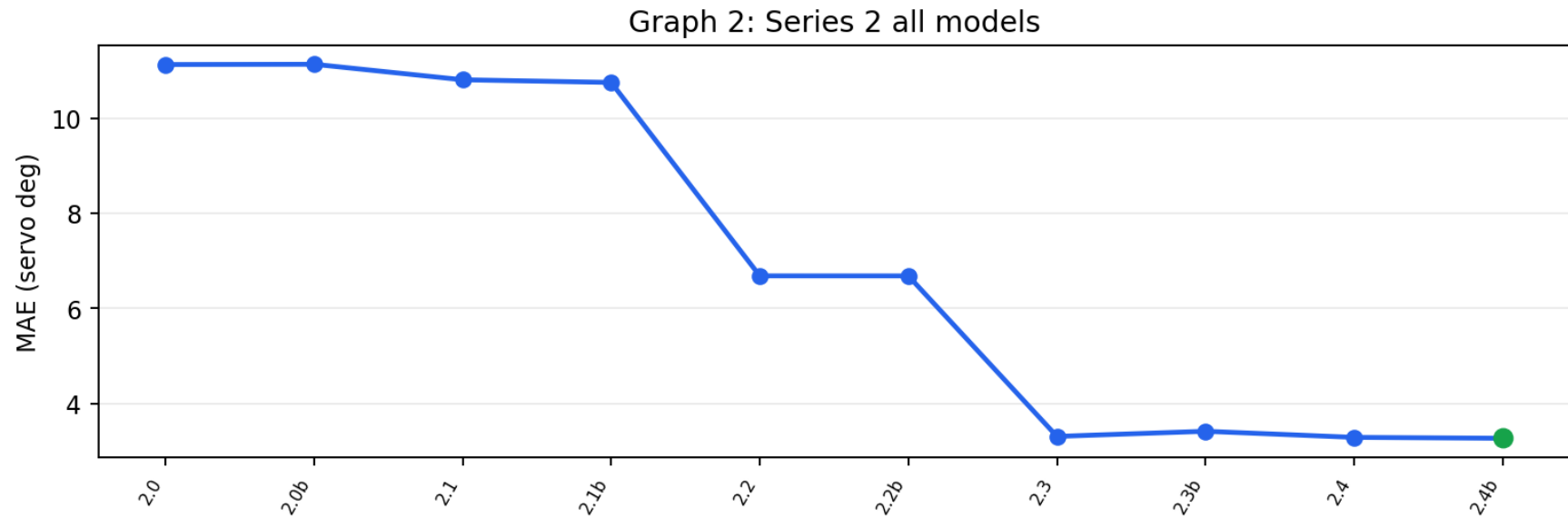
Model	HL 0-45	L 45-75	SL 75-85	ST 85-95	SR 95-105	R 105-135	HR 135-180	Exact	Off by <=1
1.3	88	365	310	431	399	523	108	30.8%	73.6%
1.2b	93	395	272	433	425	483	123	31.7%	74.2%
1.2	92	382	283	427	424	497	119	32.6%	74.4%
1.1	119	424	327	420	344	449	141	27.7%	68.4%
1.1b	119	424	327	420	344	449	141	27.7%	68.4%
1.0b	121	530	308	382	329	432	122	21.1%	58.2%
1.0	134	528	327	344	327	442	122	19.5%	56.2%

Graphs

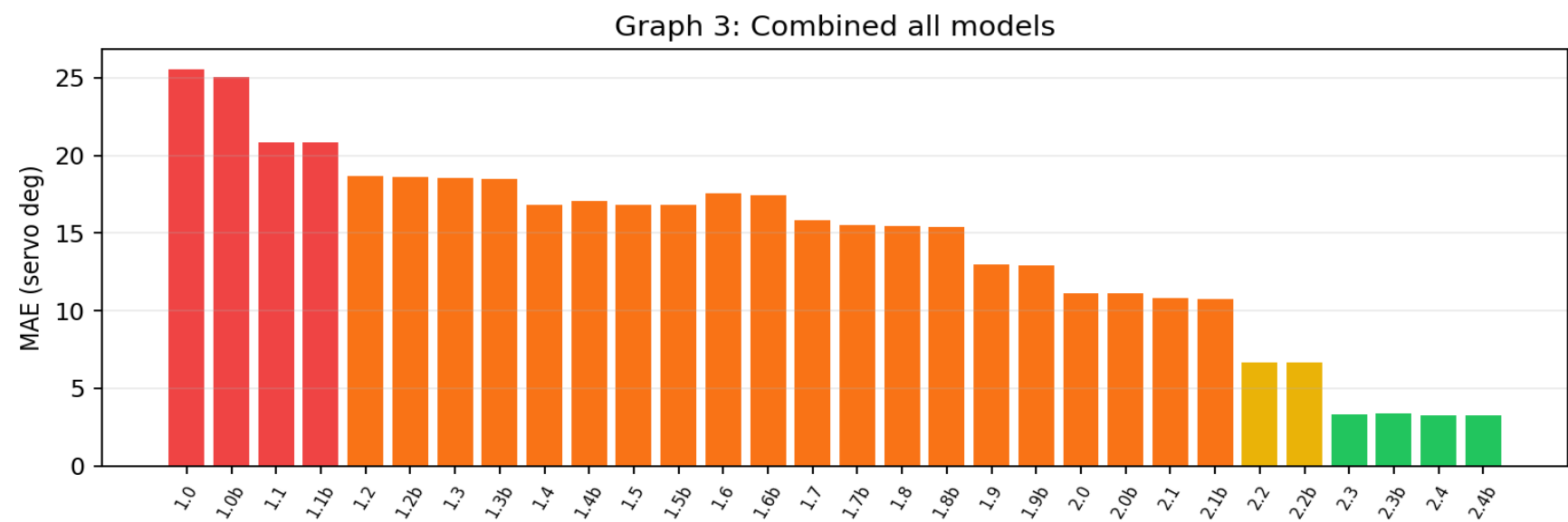
Graph 1: Series 1 all models



Graph 2: Series 2 all models



Graph 3: Combined all models



Notes

- The combined table is useful for model-selection decisions, but Series 2 v2.0/v2.0b used different preprocessing from the raw-BGR models.
- The current active label set includes D0510 field-failure images, so these numbers are not directly comparable to older 1,464-label reports.
- Offline MAE does not prove real-world reliability. The car can still fail on lighting, turns, driveways, road-edge ambiguity, speed, and sensor conditions.
- v2.4 and v2.4b were trained with the D0510 correction data; low same-dataset error may include train/evaluation overlap and should be treated as a fit check, not final field proof.